

<https://doi.org/10.1038/s44271-024-00120-6>

An Integrated theory of false insights and beliefs under psychedelics



H. T. McGovern^{1,2} , H. J. Grimmer¹, M. K. Doss³, B. T. Hutchinson⁴, C. Timmermann⁵, A. Lyon⁶, P. R. Corlett^{7,8} & R. E. Laukkonen⁹

Psychedelics are recognised for their potential to re-orient beliefs. We propose a model of how psychedelics can, in some cases, lead to false insights and thus false beliefs. We first review experimental work on laboratory-based false insights and false memories. We then connect this to insights and belief formation under psychedelics using the active inference framework. We propose that subjective and brain-based alterations caused by psychedelics increases the quantity and subjective intensity of insights and thence beliefs, including false ones. We offer directions for future research in minimising the risk of false and potentially harmful beliefs arising from psychedelics. Ultimately, knowing how psychedelics may facilitate false insights and beliefs is crucial if we are to optimally leverage their therapeutic potential.

When an idea or problem solution appears, it can produce a distinct and powerful phenomenology—a feeling of profound understanding and truth known as an *insight moment*^{1–3}. This largely ineffable experience appears to play a pivotal role in the development and adjustment of beliefs and although often associated with verifiably correct discoveries and adaptive personal growth^{4,5}, insight phenomenology can be triggered by unrelated or misleading information^{6,7} and for objectively incorrect problem solutions⁸. Insight moments are also a defining feature of psychedelic experiences^{9,10}, and could underlie the profound belief changes seen during and after psychedelic drug use^{11,12}. Thus, psychedelics are increasingly recognised for their potential to restructure maladaptive beliefs underlying mental illness^{13–15}. However, given the fallibility of insight moments, how can we ensure that psychedelic insights align with desired outcomes, or simply with reality itself? Here, we discuss the different theoretical frameworks of insight, belief change, and the neuropharmacology of psychedelics and present an integrated model for how psychedelics can engender false or maladaptive insights, which have not yet been addressed in the literature — which we term False Insights and Beliefs Under Psychedelics (FIBUS).

Insight and the psychedelic experience

Insight, defined as the sudden appearance of a problem solution in consciousness, has a long history in psychological research^{1,16–21}. Insight moments—also known as “Aha!” moments—are a special type of problem-solving process where a problem-solver achieves a sudden and complete

mental restructuring of a problem^{1–3} accompanied by a distinct rush of satisfaction, surprise, and confidence^{22–26}. A substantial literature suggests that insights are often associated with correct solutions to problems, at least when using constrained and artificial stimuli^{27–30}. Insight experiences have been observed in several recent studies on recreational psychedelic use and psychedelic assisted therapy^{31–34}, suggesting that subjective experiences of insight play a key role in psychedelic assisted therapy^{9,35,36}. These findings have garnered excitement in the field of psychedelic research, as insight moments have a long tradition of research and have generally been found to predict accuracy in problem-solving experiments^{22,29}. Indeed, psychedelic assisted therapy trials have reported many instances of insight moments during psychedelic experiences precipitating positive changes to mind and behaviour such as smoking cessation^{9,31,37–40}, potentially making them a crucial lever for clinical improvement^{32,41,42}.

The eureka heuristic

This varied body of evidence, therefore, generally supports a link between insight moments and what we deem to be “true”, either in the narrow sense of objectively correct problem solutions or broader, idiosyncratic changes in perspective and beliefs that are “verified” by their connection to positive outcomes, relative to one’s goals or values. However, recent research has shown that this link may be explained as a metacognitive process wherein insight phenomenology guides the selection of ideas—the “truth” of which depends more critically on one’s prior information. Laukkonen and

¹School of Psychology, The University of Queensland, Brisbane, QLD, Australia. ²The Cairnmillar Institute, Melbourne, VIC, Australia. ³Department of Psychiatry and Behavioral Sciences, Center for Psychedelic Research & Therapy, The University of Texas at Austin Dell Medical School, Austin, TX, USA. ⁴Faculty of Behavioural and Movement Sciences, Cognitive Psychology, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands. ⁵Division of Psychiatry, Department of Brain Sciences, Centre for Psychedelic Research, Imperial College London, London, UK. ⁶Institute of Psychology, Leiden University, Leiden, The Netherlands. ⁷Department of Psychiatry, Yale University, New Haven, CT, USA. ⁸Wu Tsai Institute, Yale University, New Haven, CT, USA. ⁹Faculty of Health, Southern Cross University, Gold Coast, QLD, Australia. ✉e-mail: h.mcgovern@uqconnect.edu.au

colleagues⁵ (2023) propose The Eureka Heuristic—the theory that feelings of insight play a heuristic role in guiding epistemic decision-making about which ideas to trust by imbuing them with a sense of obviousness. Usually, this sense aligns with reality, as other heuristics are often grounded in statistical norms, making them a generally sensible shortcut. However, as with other heuristics, The Eureka Heuristic can fail under conditions that violate these statistical norms.

False insights and metacognitive illusions

Indeed, while often correct, a proportion of insight moments are false^{22,43} and false insights can be experimentally induced⁸. For example, Grimmer et al.⁸ had participants read a series of words with high semantic similarity (e.g., Remember, Significant, Honour, Tribute, Memorial), before solving anagrams that were visually similar to another word sharing the same semantic space to the studied list, (e.g., MEMUNOMT tends to be incorrectly solved as MONUMENT instead of MOMENTUM). Participants had more false insights for the anagrams that were visually similar (to a primed associate) compared to a range of controls. The accuracy of insight moments can thus partly depend on the relevance of one's prior information, the ease (or fluency) with which it is processed, and whether the available information to the problem-solver encourages accurate or inaccurate associations.

Another downfall of using feelings of insight as a guide for truth can be seen when insight phenomenology can be misattributed, making irrelevant facts and worldviews feel true^{6,7}. Across several studies, Laukkonen and colleagues^{6,7}, presented participants with propositions (e.g., “there is no such thing as free will”) or factual statements (e.g., “lobsters can be trained to understand verbal commands”) containing anagrams (e.g. the word “lobsters” was scrambled). When participants solved the anagram via an insight moment, the irrelevant insight was temporally associated with a worldview, and participants' belief in the worldviews and statements were stronger than when no insight moment was reported. Insight phenomenology thus appears to contaminate participants' judgements during claim evaluation.

In the episodic memory literature, there is a similar concept of misattribution that can produce memory distortions⁴⁴. Like insight feelings, fluency driven by semantic activation or repetition enhances feelings of familiarity that can be misattributed to novel stimuli, resulting in false memories^{45–48}. A parallel between the episodic memory literature and psychedelic literature is that the feeling of knowing from familiarity⁴⁹ has been referred to as “noetic consciousness,” and the undeniable sense of knowledge produced by psychedelics has been referred to as the “noetic quality”³⁶. The noetic quality has also been linked to the experiences of acquiring knowledge in a seemingly unmediated fashion during spiritual or religious experiences^{50,51}. Lastly, the noetic quality is closely linked to feelings of ‘truthiness’ as described above, or ‘obviousness’ as it is often called in insight research.

However, the impact of insights goes beyond the moment of their occurrence—they may also recursively reinforce certain beliefs. For example, an individual may ‘do their own research’ about what caused the twin-towers to collapse. An insight moment at an early stage of research that makes an unfounded claim appear true could lead an individual down unproductive research pathways, and recursively induce further misleading insights. Along similar lines, false insights can also become entrenched via unfounded plausibility^{8,52,53}. Again, parallels can be found in the episodic memory literature. Emotionally positive memories tend to engage more associative and semantic processes^{54,55} that can result in memory distortions^{56,57}, and negative valence can attenuate fluency-driven false memories⁴⁶. Intriguingly, the noetic quality tends to occur with positive affect during psychedelic-induced mystical-like states⁵⁸.

Although the interaction between positive valence and false beliefs is thought to be evolutionarily adaptive^{59,60}, Laukkonen et al.⁵ proposed these related processes may mutually reinforce each other. When a false insight occurs, the positive affect accompanying the insight affirms the other beliefs the individual has—a process through which an agent can form a model increasingly out of touch with reality⁶¹. The strong link between insight phenomenology, belief, and accuracy, may hold particular importance in

the context of psychedelic use, as many users cite a desire for knowledge and understanding in their motivation for taking psychedelics. As interest in psychedelics has grown, the mechanisms behind these phenomena have been mapped within the now dominant computational paradigm known as active inference or predictive processing, a framework that unpacks potential mechanisms of the power of insights in determining belief formation over time. Below, we discuss the broad principles of predictive processing and outline a related theory of how psychedelics change beliefs via this process, before synthesising these perspectives into a novel model of how psychedelics can induce (false) insights and (false) beliefs – which we have termed FIBUS.

Active inference: a neurocomputational understanding of insight, ideas and beliefs

An increasingly popular view conceives of the brain as an inference machine^{62–64} that infers the most likely cause of sensory data so that it can optimally infer their hidden causes (i.e., what “out there” is causing these electrical signals?) These predictions (i.e., guesses about the cause of sense data) are compared to likelihoods (i.e., clarity or precision of sense data and its weighting) to arrive at posteriors (i.e., the updated prediction about the cause of sense data following the comparison between priors and likelihoods). If sense data refute priors, this solicits a prediction error – a signal informing the brain it must update its prior with respect to the stimulus at hand, or to look elsewhere for confirmatory data (i.e., active inference⁶³). In this way, the brain seeks to continually minimise prediction error^{64,65} (see refs. 64,65 for conceptual overview).

For example, if agent A thinks that the only red fruit is an apple, they will expect that the red fruit in their hand is an apple. In other words, their prior (i.e., existing expectation) is that the red fruit-like objects are most likely to be apples. If they were given another red fruit to hold (e.g., a tomato), A may notice textural differences and be told that this is another type of red fruit. This departure between A's prior that the only red fruits are apple, and the sensory data at hand (i.e., likelihood) that tells them they are holding another type of red fruit, would solicit this prediction error signal. This prediction error signal would thus inform A's internal model of the world that the initial prior was incorrect, allowing A to change the model such that ‘red fruit’ can include apples and tomatoes. Via this process, beliefs evolve in a continuous trade-off between priors, likelihoods, and posteriors (which then inform priors). Beliefs can thus be conceived as the priors carried into each sensory encounter and are equivalent to probability distributions of possible sensory encounters, evolving alongside the agent's interactions with the world. This is the definition of belief we adopt hereafter.

Predictive processing assumes the process of prediction error minimization occurs along a hierarchical neural architecture⁶⁶. Hierarchically higher cortical regions encode complex concepts pertaining to longer timescales and higher abstraction^{66–70}. In contrast, lower-level sensory regions (e.g., visual cortex; see refs. 67,71–73) typically encode more domain-specific, concrete information pertaining to shorter timescales^{66,74}. Predictions are carried down the hierarchy from higher-level cortices (e.g., associative areas such as frontal cortex) to lower-level cortices (i.e., early columns of the visual system such as V1) to narrow the range of explanations for sense data⁶⁷. In contrast, prediction errors propagate up the hierarchy such that if a prediction error cannot be explained by the next level, where more complex abstractions are encoded, it is carried to the next level^{64,67}. This process repeats until the prediction error reaches a level where it can be sufficiently explained, and the brain updates its predictions to maintain an accurate (generative) world model. This scheme allows the brain to continually refine its model of self and world for adaptive actions that aid survival^{66,70,75}.

Precision

Underwriting much of the trade-off between priors, likelihoods and posteriors is precision, an index of how narrow each probability distribution is⁷⁶. Higher precision distributions are narrower, representing increased confidence or clarity. The precision of priors and likelihoods is traded off to arrive at a posterior distribution that appropriately weighs them^{77,78}. To illustrate,

imagine Agent X carries a strong but false prediction (i.e., high precision) that the red ball they are holding in their hand is a tomato. In this scenario, it is a cold night and X is wearing gloves. Therefore, the grainy, low-precision sensory data is incapable of updating the strong expectation that “this is a tomato,” leaving the prior intact that the red ball in X’s hand is a tomato. In another scenario, X carries a lower-precision, weaker expectation about the identity of the red ball that is met with strong sensory evidence because now it is daytime, and the gloves are off. This high precision evidence (or likelihood) suggests the red ball is an apple – making it more likely X will revise their weak prior and arrive at the posterior that the red ball is instead an apple. Here, we are referring to the fact that this process allows the agent to arrive at a precise posterior distribution over models, allowing the agent to optimally navigate its sensory landscape and thus their environment.

With respect to the brain, more generalized and coarse-grained representations (based on many observations) are thought to be encoded in domain-general cortex⁷⁰. During perception, predictions cascade down to sensory regions of cortex which encode more specified and faster-updating predictions. Prediction errors ascend this cortico-perceptual hierarchy until the error can be explained by the level above. In the event the error cannot be explained at the next level, it ascends all the way up to the most coarse-grained level such that the agent’s model is updated to account for this new contingency^{67,79}. In the example above, if we assume the agent has observed many times that red circular shapes are tomatoes, they will have a relatively stronger prediction that the red ball is a tomato and be less susceptible to updating this belief when the sensory data was unclear (assuming they do not actually find out that the red ball is not a tomato).

Greater (environmental) statistical regularities often accompany increased prior precision and should therefore be accompanied by strong contradictory empirical data to be refuted and updated. For example, a black sky almost always means it is night, and only strong evidence, such as knowledge of a solar eclipse occurring, enables one to suspend the belief that the black sky they are observing is not evidence of it being night time. The difference in outcomes to which these scenarios underscore is a process called *precision weighting*, the relative weighting of priors and likelihoods during the perceptual process, given the context and prior learning. The clarity of sensory data and the relative confidence in expectations thus play a crucial role in how beliefs evolve. Reliance on sense data or likelihoods, based on their precision, is crucial for determining which beliefs are ultimately reached, with the higher-precision distribution typically being more influential.

Insight, beliefs, and active inference

Friston et al.⁶³ posited an account of insight experiences nested in the active inference and predictive processing frameworks. Under this view, refinement or modification of one’s generative model (i.e., world model) need not rely on new information—a process deemed *fact-free learning*. Fact-free learning occurs via a process of Bayesian model reduction, wherein the brain arrives at models providing more parsimony of sense data already accrued, rather than continued sampling. Fact-free learning is said to be metaphorically similar to the way that a “sculpture is revealed by the artful removal of stone”⁶³.

Such learning is proposed to occur implicitly via simplification of one’s model, in states where people are not actively taking in sensory information, such as in sleep⁸⁰, or states of interoceptive reflection⁸¹. The key point is that via Bayesian model reduction, no further sensory sampling is necessary for refining and updating beliefs about the matter at hand.

An extension to Friston et al.’s⁶³ model of insight has been proposed that considers the experiential quality of insight and its effects higher in the cortical hierarchy⁵. According to the ‘Eureka Heuristic’, we experience feelings of insight because they help ‘highlight’ which ideas we should trust in light of past learning. In other words, insight moments play a role in heuristically selecting ideas from the stream of consciousness by capturing attention, inducing confidence, and boosting drive to act on them. In an uncertain world where time is limited, ideas cannot always be evaluated analytically. The feeling of insight plays a key role in permitting quick and efficient action on novel ideas (e.g., when running from a lion on the savannah).

This view is consistent with work suggesting that insight moments increase confidence^{25,26}, can be misattributed^{6,7}, lead to beliefs that are difficult to forget⁸², and are resistant to revision²⁸. In some ways, insights can be considered a fast-track to semantic memory, bypassing the slow training process between the hippocampus and cortex that typically give rise to semantic memories imbued with noetic consciousness. The Eureka Heuristic also includes computational mechanisms that extend the Bayesian reduction account above, and helps us understand the recursive, reinforcing role that insights may play during a psychedelic experience. We summarise the model below.

The Eureka Heuristic proposes that when the implicit process of Bayesian reduction results in a novel (i.e., updated) model, it necessarily solicits a prediction error at a higher-order conscious level of abstraction, under the hypothesis that precisely held beliefs enter consciousness. This prediction error in turn can change one’s model at a conscious level, making it possible to have a reportable insight (and meta-awareness of an insight having occurred). In other words, while Bayesian reduction inherently reduces global prediction errors across the system, a certain amount of time is required to share this information via ascending prediction errors.

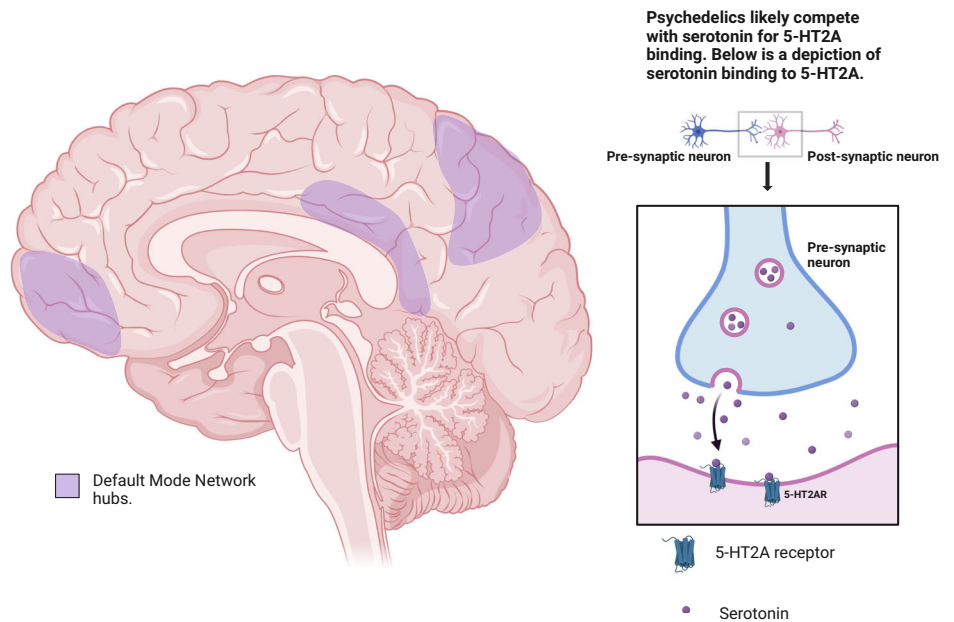
Crucially, since many ideas can appear in the mind, it is only the ideas that have high expected precision (i.e., subjective confidence in the ideas)—thus feeling insightful—which are selected and meaningfully impact beliefs. This is analogous to the way that an organism must infer both the action policy and (dopaminergic) confidence in it^{63,83}. Similarly, organisms infer both the content of ideas as well as their (dopaminergic) insightfulness.

Prediction errors ascending the cortical hierarchy can be ascribed higher or lower precision weighting, as can predictions that descend the cortical hierarchy (although it is worth noting that this may not always be the case, as lower level prediction errors may not invariably propagate to higher levels if the loss of accuracy is accompanied by complexity reduction). Notably, insight experiences have all the neural characteristics of a higher-order prediction error (e.g., restructuring and insight is associated with the event related potential component N320^{84–86}). More precise prediction errors (i.e., very sharp departures from predictions with strong evidence) are thought to enact a larger influence in (Bayesian) belief updating, such that they bear increased weighting compared to the agent’s priors. Increasing the precision weighting of prediction errors is thought to be instantiated by higher synaptic gain (i.e., the inhibitory or excitatory strength of connections between neurons)^{5,87,88}. One way synaptic gain is instantiated is the up-regulation of dopamine, through which belief updates and thus confidence in belief updates are thought to occur^{87–90}.

Like the construct of precision, the feeling of insight is thought to be implemented through dopamine and has been linked to the reward system^{91,92}. Just like precision, insight experiences are associated with attentional capture^{30,93}, higher confidence and (phenomenological) pleasure^{22,25}, and seem to map onto the dopaminergic reward system^{91,92}. Moreover, in contrast to norepinephrine, which is thought to retain the dependency of episodic memories on the hippocampus, dopamine is thought to facilitate the integration of episodic memories into cortical semantic networks⁹⁴. Precision also drives model selection, just like insight drives the selection of new ideas^{5,12}. Thus, what we call ‘insight experiences’ map extraordinarily well to the computational construct of a precise prediction error at an abstract level. Put simply, insights are a surprising inner event (prediction error) imbued with noeticism given what one knows (high expected precision), thus permitting idea selection and action.

We note that dual process theories provide a framework for understanding cognition as a binary of ‘conscious, deliberate, effortful’ and ‘unconscious, rapid, and largely involuntary’ thought⁹⁵. However, more recent work has expanded upon this concept and empirical findings (such as the phenomenology of insight occurring in traditionally analytic problems⁹⁶) and identified the need for a more comprehensive view of cognition as occurring as a hierarchy, with “system 1” and “system 2” effectively existing on opposite ends of a continuum encompassing all of conscious thought, unconscious judgement and decision-making, as well as even ‘lower-order processes’ such as perception and emotion⁹⁴. Indeed, insight can occur

Fig. 1 | Depiction of the neuropharmacology of psychedelic effects on the brain. Cortical regions that comprise the DMN (medial prefrontal cortex, posterior cingulate cortex, angular gyrus, and precuneus) are shaded in purple. These DMN regions include the densest expression of 5-HT_{2A} receptors, which psychedelic drugs bind to, resulting in disrupted functioning of the DMN. We note this is a simplified portrait (see de Vos et al.¹⁰⁵ for a detailed overview) (created with BioRender.com).



across both types of thinking^{25,26,81,97,98}. One of the key deviations that PP takes from these earlier theories is it emphasises these lower-level processes as occurring mostly prior to (outside of) conscious awareness. This is also a key component of our argument, as predictive processing can be applied to phenomena that have until now been thought of as completely “conscious” such as belief, insight, and even cognitive dissonance theory.

Psychedelics and belief change: two possible pathways

We have thus far covered the notion of insight and the key role that it can play in updating beliefs via predictive processing. Similarly, predictive processing has been suggested to explain belief change under psychedelics in an influential theory known as Relaxed Beliefs Under psychedelics or REBUS⁹⁹. We discuss how REBUS effects from psychedelics can result in belief changes. Following the description of REBUS and how it can drive belief change, we then describe an alternative pathway to psychedelic-facilitated belief change that does not rely on the assumptions of REBUS. After introducing these relevant theories, we will then present our integrative account for how belief change under psychedelics can engender false beliefs drawing upon components of each.

REBUS and belief change

Psychedelic substances, such as LSD, psilocybin, and DMT, primarily act as agonists at the brain’s 5-HT_{2A} serotonin receptors, which are widely distributed throughout the brain, particularly in regions associated with high-level cognition, such as the prefrontal cortex, and sensory processing, like the visual cortex¹⁰⁰. When psychedelics bind to 5-HT_{2A} receptors, they cause increased excitation of neurons, leading to altered patterns of neural activity and communication. This heightened excitation is thought to contribute to the profound perceptual, cognitive, and emotional effects of psychedelics¹⁰¹. In addition to their actions on 5-HT_{2A} receptors, psychedelics can also influence other neurotransmitter systems, such as dopamine and glutamate, which further modulate neural activity and contribute to their complex effects¹⁰². Neuroimaging studies have shown that psychedelics induce changes in brain connectivity, reducing the connectivity of the default mode network (DMN), a group of brain regions involved in self-referential processing and inner thought^{103,104}. This disruption of the DMN is hypothesized to underlie the “ego dissolution” and sense of unity often reported during psychedelic experiences^{103,105}. Simultaneously, psychedelics enhance connectivity between other brain networks, potentially facilitating novel associations, insights, and perspectives. The combination of receptor-level

effects, neurotransmitter modulation, and large-scale network changes induced by psychedelics is thought to create a unique brain state that supports profound alterations in consciousness, perception, and cognition^{99,101}. Below, Fig. 1 provides an overview of the neuropharmacology of hallucinogenic substances mode of action.

REBUS proposes that psychedelics facilitate belief change via a two-step process⁹⁹ (Carhart-Harris & Friston, 2019). First, psychedelics disproportionately diminish the precision weighting of high-level priors (e.g., reducing confidence in ‘beliefs’ in the colloquial sense, formalised as higher variance probability distributions) that otherwise constrain lower levels (e.g., perception). This assumption of disproportionate higher-level effects is due to the densest distribution of 5-HT_{2A} receptors found in certain association cortices, including parts of the default mode network (DMN), proposed to be the top of the brain’s hierarchy (note, however, that 5-HT_{2A} receptors are also densely distributed in visual and auditory cortices¹⁰⁶). By relaxing the DMN’s constraints on the rest of the brain, the brain’s hierarchy of information processing is thought to be “flattened” or less controlled and constrained by higher-order abstraction and ‘freer’ to change according to new input (note that by higher order, and in a more technical sense, we are referring to beliefs about plausibility’s of a set of models that are updated). A secondary consequence of the system being unable to rely on prior assumptions is the relatively increased precision weighting of sensory data, resulting in novel input becoming more likely to impinge on high-level beliefs. An agent under psychedelics may therefore consider any number of alternative hypotheses about the causes of sensory data, perhaps rapidly, and revise higher-order beliefs that were held in a sober state. Particularly under high doses, psychedelics can produce a collapse of complex assumptions such as one’s sense of self, one’s membership to a group, and typical knowledge about the world, coinciding with the DMN’s role in self-referential processing¹⁰⁷, social processing¹⁰⁸, and semantic memory¹⁰⁹.

The REBUS hypothesis is supported by neuronal, behavioural, and clinical data. Evidence suggests that psychedelics reduce top-down connectivity and dampen the power of backward travelling waves (i.e., signature of neural activity traveling across cortex, suggesting a decreased activity between higher and lower levels in the brain)^{110,111}, both suggested mechanisms for the influence of priors on brain activity¹¹². After the acute psychedelic experience, there are documented changes to metaphysical beliefs, particularly away from a physicalist worldview (we note this does not provide evidence exclusively for REBUS- but just that psychedelics can seemingly change beliefs)^{113–115}. Finally, qualitative studies from clinical trials suggest that revision of self-related beliefs (arising from REBUS

processes) may underpin positive psychological changes^{10,40,116}. Whilst these effects may be beneficial as a metaphorical ‘reset’ if one holds an array of maladaptive beliefs, there is no guarantee that relaxing one’s hard-earned abstract understanding results in positive change.

Alternate pathways to belief revision under psychedelics

Outside of the active inference or computational frameworks, psychedelics may impact beliefs via effects on fluency and relative weighting of hippocampal and cortically dependent memories. This pathway to belief revision, which we will term a ‘memory systems account’ does not preclude REBUS effects, but we highlight this account since there may be differing predictions on the specific brain-based substrates to belief changes.

Feelings of insight and familiarity can come from fluency manipulations such as semantic priming (e.g., seeing the word *whisker* could activate the category of *cat*)^{8,45}, which can be enhanced by psilocybin¹¹⁷. Moreover, although psychedelics impair the formation of hippocampally-dependent recollection memories (e.g., remembering/recollecting where or when an event took place), they spare or even enhance formation of cortically dependent memories that solicit feelings of familiarity (e.g., knowing a face, without who the individual was or where they met them¹¹⁸).

Typically, hippocampal recollection may constrain the interpretation of noetic feelings driven by fluency/familiarity. If one can explicitly recall semantically relevant words or the multiple repetitions of a stimulus’ presentation, they may be better able to understand the source of their noeticism and not misattribute it to irrelevant stimuli. For example, a person might continually observe they are in a friend’s house one evening and can thus attribute this fact to explaining why they keep remembering the presence of their friend, instead of mistakenly attributing this feeling to the fact they saw their friendship bracelet that reminded them of their friend. In contrast, non-drug studies have found that when recollection fails and familiarity is high, *presque vu* (illusory feelings of insight) can emerge¹¹⁹, as well as other phenomena sometimes reported under psychedelics such as *déjà vu*¹²⁰ and premonition¹²¹.

In models of memory systems, the hippocampus is thought to “train” the cortex over time such that greater statistical regularities between episodic memories are what eventually become semi-permanent semantic memories (e.g., one may no longer have memory for every instance they had pizza or even the first time they had pizza, but they have learned what a pizza is)¹²². The hippocampus may even constrain what the cortex can learn by providing contextual information that biases cortical processing¹²³. Some work suggests that conditions in which hippocampal activity is relatively disconnected from the cortex such as during rapid eye movement sleep is important to the instantiation of new cortical information¹²⁴. High-level beliefs can be thought of as semantic memories not necessarily shared by others (e.g., “I am a bad person”), as they are typically slowly learned over time, difficult to revise (it would be hard to forget what a pizza is) and represented by association cortices including the DMN but especially the anterior temporal lobe¹²⁵. In fact, the anterior temporal lobe is an important site for insight learning^{93,126,127}, familiarity^{128–130}, semantic priming¹³¹, the illusory truth effect¹³², and the formation of beliefs such as prejudice¹³³.

By reducing the constraints of recent hippocampal memory (i.e., impairments of forming recollections) via inhibitory 5-HT_{2A} receptors in entorhinal cortex (i.e., the input to the hippocampus) and the hippocampus itself^{134–137} and facilitating cortical processing (i.e., fluency) via excitatory 5-HT_{2A} receptors in the cortex, psychedelics may be able to revise semantic stores supporting high-level beliefs. Less constraints may provide greater exploration of a conceptual search space allowing one to reach veridical insights. However, noetic feelings arising from aberrant semantic activation could also be misattributed to unrelated or bizarre ideas produced by psychedelics, resulting in false insights.

REBUS proposes that the hippocampus is one of the regions that becomes less constrained by the cortex under psychedelics and thus should increase its influence on the cortex, especially the DMN⁹⁹. In contrast, this memory systems account predicts that typically the cortex is constrained by the hippocampus and that under psychedelics, the cortex becomes free of

such constraints. It has been found that psilocybin attenuates hippocampal-DMN coupling¹³⁸ and hippocampal glutamate, which is predictive of *feelings* of insight, but not necessarily veridical insights¹³⁹. Nonetheless, all accounts converge on the general notion that psychedelics change beliefs, even if mechanisms are debated. We now turn our attention back to insight. Crucially, we suggest that it may play a key role in entrenchment of new beliefs following psychedelics.

Psychedelic-induced insights: a possible pathway to false beliefs

Considering the theories of insight and belief change under psychedelics discussed thus far, we now turn to the pressing issue identified at the outset—the possibility that psychedelics could engender false beliefs. Although psychedelics show promise as tools for engendering insight and therapeutic belief change, the neurocomputational perspective on insight and belief change in general suggests that psychedelics could also elicit false beliefs under some circumstances. For instance, psychedelics could merely increase the frequency of belief changes, orthogonal to accuracy, with the utility of these belief changes depending on the accuracy of one’s prior information at the time of restructuring. Give the evidence that insight moments can often be wrong or misleading due to cognitive or environmental factors^{4,8,52,53,140}, a higher frequency of insights (both true and false) could also increase the probability of psychedelic-induced maladaptive, or potentially false, insights¹¹. We now sketch a candidate framework for how psychedelics engender belief changes *via* soliciting insight moments.

Our proposal is as follows: Psychedelics imbue a decreased ability to make sense of sensory data, leading to an increased number of insight moments and noetic feelings. Following the experience, the person may be left with a lack of detailed memory, but an increased noetic confidence in the insight moments encountered during the experience. Crucially, the increased quantity of insights and acute malleability may leave one vulnerable to empirically false, misleading, or maladaptive insights, alongside the prospect of obtaining valuable new perspectives.

Note that our focus on false beliefs is not because we believe that psychedelics only solicit incorrect ideas, but because the potential for false beliefs under psychedelics have been somewhat overlooked¹¹. Secondly, if psychedelics do hold potential to change deeply held beliefs—as they are believed to⁹⁹—and some proportion of these are likely to be false but feel profoundly true and motivating, there are major consequences to consider. Given the renaissance that is currently underway, mass adoption of psychedelic use both clinically and beyond have an important epistemic task to address: how do we improve the likelihood that the insights and subsequent belief changes engendered by psychedelics result in beliefs that move one closer to reality? Below, Box 1 provides a summary of the similarities and differences between these models, including our proposed model.

False insights and beliefs under psychedelics (FIBUS): towards a theoretical account

Based on our prior discussion of precision, model reduction, and fact-free learning, we propose a process for how insight-derived belief changes under psychedelics may reorganise belief structures – which we have coined FIBUS. We do not just account for the mechanism of insight and belief change under psychedelics, but also articulate how insights can be true or false, with clear implications for future studies (see below) involving psychedelics. We propose this process approximates four steps as follows, drawing on all the research thus-far reviewed.

Figure 2 illustrates the process by which psychedelic induced insights may engender false beliefs. First, increased agonism of serotonin 5-HT_{2A} receptors results in decreased precision weighting of priors, such that priors now no longer characteristically constrain perception and cognition (See^{141–143} for in-depth discussion on serotonin, dopamine, and precision). Second, this collapse in the perceptual-belief landscape results in novel thoughts and perceptions that then increase the incidence of prediction errors (e.g., through new sensory input or via ‘fact-free learning’). These prediction errors facilitate new ways of interpreting sensory data and

Box 1 | Outline of the overlap and departure between REBUS model and the model of psychedelic belief change forwarded here: FIBUS. Overlap between REBUS and FIBUS is bolded, and differences in the FIBUS model are non-bolded

REBUS	FIBUS
• Psychedelics decrease precision weighting of priors via their diminishing of the default mode network. • This diminishing of the DMN results in the brains hierarchy being ‘flattened’, resulting in altered perceptions and inferences in a more flexible system. • Psychedelics can change beliefs because of the novel information being integrated, and because of the newfound flexibility afforded to the brain.	• Psychedelics decrease precision weighting of priors via their diminishing of the default mode network. • This diminishing of the DMN results in the brains hierarchy being ‘flattened’, resulting in altered perceptions and inferences and is more flexible. • Due to this flexibility the brain can arrive several new insights, and the altered and number of unusual perceptions increases. Psychedelics thus imbue a decreased ability to make sense of sensory data, leading to an increased number of insight moments and noetic feelings. • The trade-off in the precision of priors (decreased) and sensory data (increased) results in a heightened number of insight moments during the acute psychedelic experience, increasing dopamine. The insights and predictions errors are afforded higher precision due in part to the fact that priors are now diminished which in turn makes insights more likely to influence subsequent beliefs. Concurrently, increased fluency, and weighting of hippocampal and cortically dependent memories under psychedelics result in diminished constraint of the hippocampus, enabling increased flexibility to the cortex (see section on the Eureka Heuristic, and Alternate Pathways to Belief Revision Under Psychedelics for detail). • Psychedelics can change beliefs because of the novel information being integrated, and because of the newfound flexibility afforded to the brain. • Crucially, nothing about psychedelics preferentially selects for accuracy, but just the feelings of accuracy. This leaves the system vulnerable to misleading contextual information wherein one can feasibly feel confident in beliefs and insights arrived at under misleading contextual circumstances or cognitions.

generate new ideas and perspectives. Third, the decreased precision weighting of prior beliefs affords increased precision to the novel input passed to higher levels (i.e., *everything feels more insightful* because it is not constrained by prior belief). This increased precision weighting is thought to be implemented via dopaminergic release (note this is a secondary release not directly facilitated by drug effects; see^{5,87,88,91,92,144}), affording higher precision to the insights encountered in step two. Fourth, this increased precision weighting given to the insights makes them more likely to feature in model selection (i.e., the feeling of insight has an unusually strong effect on belief updating).

This process, we suggest, can describe how beliefs arising from psychedelic insights can become entrenched in working models thereafter. Such entrenchment may be especially true for psychedelics that also activate dopamine receptors such as LSD, which tends to have more lingering effects on perception¹⁴⁵. Critically, we suggest that while this process can impart adaptive, meaningful, and lasting belief changes, it can also facilitate *false* insights (and hence false beliefs). Below, we discuss implications for this model, with a particular eye toward how practitioners and researchers alike may consider, test, and optimally refine these dynamics to ensure optimal treatment protocol.

A model of psychedelic insight and belief change, and its implications

We suggest that psychedelics can provide a genesis for false beliefs as follows. First, REBUS effects (or other mechanisms of decreased precision weighting) may induce belief relaxation, including those that are true, and increase precision weighting of novel dopaminergically modulated insights. The insights afforded by increased precision subsequently bear disproportionate weighting on model building. In some instances, ‘fact-free’ learning may therefore be occurring primarily with respect to erroneous or embellished

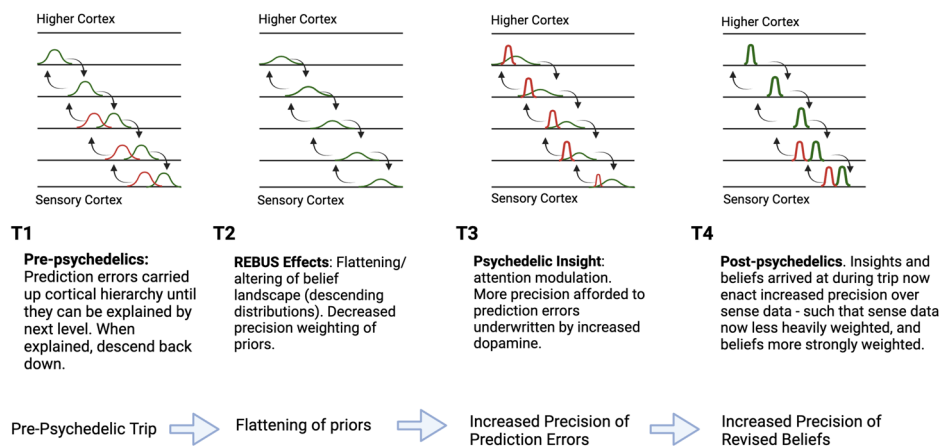
sensory data. Moreover, the insights may then play a recursive role of preferencing and entrenching ideas consistent with the new beliefs. Concurrently, the impairment of hippocampally modulated recollection may lead to a decreased ability to remember veridical details of the experience, while cortically facilitated memory encoding leads to increased semantic aberrance and noeticism. In the experience, one encounters 1) an impaired apparatus to make sense of incoming sensory data, 2) increased insight moments, and 3) increased feelings of familiarity irrespective of accuracy. After the experience, one is left with a lack of detail of memory, but an enhanced noetic sense about the insight moments of the experience. Psychedelics may thus facilitate insights and increase the *perceived* novelty in new ideas and original thoughts^{11,139}. However, nowhere does this experience preference accuracy, or a necessary nudge toward more adaptive beliefs characteristic of improved mental health, leading us to describe this process as one of False Insights and Beliefs Under psychedelics (FIBUS). Below, we outline considerations that accompany this empirically derived model and proposal. We divide our considerations between the acute and post-acute phases.

Acute effects

A potential downside of (higher order) belief relaxation is that some adaptive priors that typically constrain perceptual inferences may also be relaxed in the process leading to false insights and hence false beliefs. This amplifies the oft-cited importance of set and setting, which are crucial predictors of the psychedelic experience¹⁴⁶. With respect to setting, misleading contextual information could serve to increase the possibility of false insights. Psychedelics may serve as amplifiers of environmental influence rather than pushing someone toward one set of views over another¹⁴⁷. A recent study found that changes in metaphysical beliefs following a psychedelic experience were mediated by a range of factors (i.e., age, personality

Fig. 2 | Depiction of REBUS effects and how they result in insight moments and belief changes (Figure created with BioRender.com). The red

distributions represent prediction errors, and green distributions represent predictions. Higher levels encode more domain general, complex abstractions (i.e., high-level beliefs). Lower levels, such as the sensory cortex, encode simpler and domain-specific concepts (i.e., low levels, such as edge detectors¹⁷²). High levels send descending predictions, while prediction errors ascend the hierarchy until they are explained away by higher levels. Insights (i.e., prediction errors) arrived at during the trip are afforded higher precision due to decreased precision weighting of priors, rendering insights and consequent beliefs higher precision thereafter — a process that we have termed FIBUS.



traits, suggestibility) reinforcing the notion that non-pharmacological factors play an important role in adopting novel beliefs^{11,115}.

Aspects typically emphasised in the clinical setting, such as safety and control, may additionally provide patients overly precise priors of felt safety or control in a non-clinical setting, where psychedelic consumption may not be safe nor well controlled. If psychedelics do acutely enhance fluency (ease of retrieval in memory¹⁴⁸), this may result in an exaggerated mere exposure effect in which patients become more attached to those they are interacting with whilst under psychedelics. There might also be aspects of the clinical setting conferring discomfort or distrust of clinicians, making future care more difficult. Of course, this could simply mean that any aspect of the experience that is out of step with day-to-day life could be overweighted, and thus garner outsized influence following the acute phase – consistent with the neutral amplifier of set and setting discussed above. Out of the psychedelic context, insights and beliefs arrived at may no longer carry their adaptive zeal. This amplifies the importance of an epistemically congruent (i.e., with the goals of the patients who ingests the psychedelic) set and setting – given that beliefs become more malleable during the acute phase, contextual factors in the post-acute phases (in which ‘integration’ therapy occurs) need to optimally encourage and support positive and adaptive insights.

We also note there may exist theoretical shortcomings to the REBUS model, which can paint an incomplete picture of belief change, and thus constrain our FIBUS model given we derive our predictions partly from the REBUS model. Some psychedelic studies find larger effect sizes outside of associative cortex, including in sensory areas with lower 5-HT_{2A} distribution^{102,104,149,150}. Another outstanding question on REBUS effects is the presence of hallucinations during the acute phase. People often report reliving scenes from one’s past and immersive hallucinations of beings or ‘entities’, often considered high level-hallucinations. If the sense of self collapses under psychedelics, then it is unclear how one could have hallucinations that assume a sense of self (e.g., I am having a memory of something I have experienced before). Given our FIBUS model partially derives from and assumes REBUS effects, these shortcomings should be noted.

Another consideration with respect to our FIBUS model is the suggestion that hallucinations are due to overly strong priors – such as in acute episodes of psychosis¹⁵¹. If complex hallucinations are occasioned by strong priors, then we might expect *less* complex hallucinations at higher doses. Indeed, complex hallucinations are typically only occasioned with high doses and sensory deprivation (for example, in the case of Charles Bonnet syndrome^{151,152}). One explanatory model for hallucinations from dissociative hallucinogens (i.e., NMDA antagonists such as ketamine), which share some subjective, clinical, and neural effects as psychedelics, proposes the opposite of REBUS¹⁵³. That is, hallucinations come from an overweighting of priors (e.g., “I am seeing my mother”) and an underweighting of sensory information (e.g., external input that would otherwise lead one to

reject the idea that their mother is present). Indeed, the intensity and complexity of the hallucination (e.g., whether it is just geometric shapes, or reliving complex past experiences) could be highly dependent on dose, as well as effects on other cortical regions.

Recent work suggests that the degree of belief changes rests on how strong the prior initially was^{154,155}. As such, weaker priors may be further weakened (e.g., psychedelics shifting a slightly differing opinion closer to the social norm¹⁵⁶), and stronger priors may become stronger due to the sociocultural and local environment tipping cognitive systems toward one belief (e.g., political liberalism) or another (e.g., political conservatism) during and after the acute phase¹⁵⁷. Indeed, the current evidence seems to suggest that higher-level beliefs may be susceptible to change, although environmental noise (e.g., local, and broader sociocultural setting) may act as mediators of belief change as well^{115,155,157}. Whilst these concerns do not preclude the FIBUS proposal here, future work should aim to further investigate the relationship between hallucinations, mechanisms pertaining to REBUS, and belief alterations.

Post-acute effects

The dopaminergic surge accompanying insights, combined with the memory alterations, may result in an undue sense of confidence for insights accrued. Insights gleaned during psychedelic experiences may therefore bear increased weighting in model selection (i.e., the set of beliefs that make up the world model following the acute phase). For example, mystical experiences encountered during the acute phase, and insights they incur may be primary mediators of beneficial belief updates^{4,9,32,158–160}. However, it must also be noted that the subjective feeling of insight is not the same thing as a genuine breakthrough, as even mundane ideas engendered by the experience can seem more meaningful than what they really are^{99,139,161,162}. For example, Mason et al¹³⁹ found that higher decreases in functional connectivity within the default mode network predicted increased feelings of insightfulness, but decreases in objective originality. The key point is that consistent with our FIBUS proposal, insights seem to be exaggerated both in quality (subjectively defined) and in quantity during psychedelics, and they can deeply impact belief updating (perhaps even personality change¹⁶³), and thus subsequent model selection.

A second-order consequence is that these insights and subsequent beliefs may be more difficult to revise, particularly when ascribed the (memory systems modulated) noetic feelings accompanying them. A potential by-product of the precision (and associated noetic feelings modulated by memory systems¹⁴⁸) afforded to insights is the resulting belief updates may be less amenable to change^{5,28}. With psychedelic occasioned insights, participants report a non-specific feeling of truth associated with insights¹⁶⁴. If it is not clear *why* an insight feels true, it can be difficult to revise since it is not clear what information could contradict it. For example, if I have the insight that I am possessed by a negative entity, the prospect of

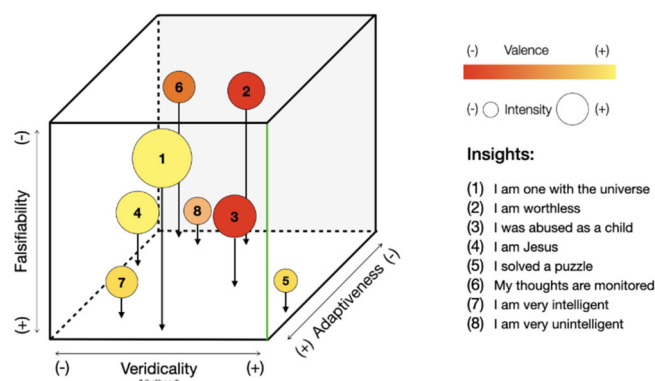


Fig. 3 | Schemata of possible insights one might arrive at under psychedelics. The colored circles are illustrative examples only, and all these quantities and qualities may vary from person to person. Each example insight in the box has a number which corresponds to a particular insight under the 'Insights' title on the right-hand side. The green axis refers to the direction which may be the overall ideal therapeutic direction (i.e., veridical, adaptive, and, where possible, falsifiable beliefs) although there are likely exceptions. Each of these dimensions is explained in the box below with reference to the examples shown. Notably, some insights may confer a sense of wellbeing (i.e., adaptiveness), but are inherently difficult to verify. As such, the challenge for a clinician may be to unpack each insight along the falsifiability-veridicality-adaptiveness axis, making sure that insights are optimally leveraged to facilitate clinical improvements.

which is central to some shamanic traditions, it may be extremely difficult to revise because the very foundation for the idea is simply the feeling. These insights seem to have lasting behavioural effects, as discussed, such as reductions in depression symptoms^{10,41}, cessation of smoking and substance use^{165,166}. This is important in a clinical context if the goal is to increase psychological flexibility such that more adaptive beliefs might be considered and adopted.

A recently developed framework aimed at addressing issues of psychedelic-induced false insights proposed the fostering of a 'gentle touch' for revelations occurring during psychedelic therapy sessions. In this framework (deemed 'psychedelic apprenticeship'), the relational processes (e.g., therapeutic interventions performed by a therapist) occurring before, during, and after the psychedelic session could serve as a scaffold for users' to hold novel insights lightly¹¹. These relational processes could be seen as a form of 'thinking through other minds' or 'cultural affordances'¹⁶⁷, whereby an experienced facilitator or therapist can aid in the modulation of the users' precision weighting of newly acquired insights or beliefs during psychedelic therapy. With respect to our FIBUS model, the gentle touch framework, alongside a rubric for making sense of the adaptiveness, veracity, and falsifiability of psychedelically derived insights (Fig. 3), could offer clinicians a framework for integrating psychedelic insights in a clinically useful way. Below, we divide discussion of our FIBUS model into research and clinical implications, offering novel hypotheses and considerations for clinicians involved in integrating psychedelic insights.

Future directions, insights, and epistemic hygiene

Our FIBUS model lends itself to novel empirical predictions. First, FIBUS predicts that subjective veracity and perceived number of insights under psychedelics should linearly increase alongside dopaminergic release during the acute psychedelic phase. This could be tested by leveraging tools such as Positron Emission Tomography (PET)¹⁶⁸ or blood sampling in the acute psychedelic phase. In the post-acute phase, researchers could ask how frequently and strongly participants experienced these insight moments during the acute phase. A second prediction here is that psychedelics should induce more false insights than during ordinary cognition, as psychedelics can increase the perceived (but not necessarily the objective) novelty or accuracy of new ideas^{4,139,169}. To test this, future researchers could have participants perform tasks whilst under varied doses of psychedelics and see

whether the number insights preceding a solution increase in number, and whether they are less accurate than during ordinary cognition. To test for strength of belief accuracy, researchers could administer, for example, a Brown Assessment of Beliefs Scale¹⁷⁰ post acutely. Of course, testing for the number and subjective veracity in these insight moments should themselves predict the strength of beliefs, without having to test for dopamine. The implication is that these beliefs should be stronger but less accurate when there was higher dopamine, and more false insights (both true and false), arising during the psychedelic experience.

A clinically relevant issue requiring elaboration is the relationship between truth and adaptiveness. As we have alluded to throughout this paper, determining what counts as a "true" belief or insight is challenging in complex domains beyond problem solving. We are also not making the claim that true beliefs and adaptive beliefs are synonymous in all circumstances. A related notion here is that of falsifiability. Indeed, many meta-physical beliefs that one may adopt following psychedelics do not easily lend themselves to testing, at least not at the individual level. For example, if one adopts panpsychist views (i.e., "everything in the universe is conscious") following the acute phase, this is not a belief system that carries a clear criterion on which it may be empirically proven or disproven. As such, the challenge for the clinician may be to determine whether the newly found panpsychist views are supportive or refutative of the persons psychopathologies. Put differently, if one does adopt views that are not easy to amend or dispute, the clinical challenge becomes whether the belief betters or worsens the patients' clinical symptoms. It could well be that when objectivity and falsifiability are not able to be established, adaptiveness could become a more important locus for care. As such, the relative weighing of these factors may depend on clinical judgement and the specific clinical goals of the patient.

Optimal integration of psychedelic insights in the clinical setting will be an important aspect of psychedelic-assisted interventions. However, this is no easy task given that any one insight can vary along several dimensions including subjective intensity, emotional valence, as well as veridicality. Moreover, as touched upon above, the contents of insights may vary along dimensions including veridicality (i.e., how likely or unlikely an insight or belief is to be objectively true), adaptiveness (i.e., whether an insight conducive to better or worse clinical outcomes), and falsifiability (i.e., whether the insights and consequent beliefs easily are updatable based on sampling more sensory data). In Fig. 3, we provide a tool for thinking about this potential space of psychedelic insights, which may assist in identifying relatively more and relatively less desirable insights. This relates to the notion of 'epistemic hygiene'¹⁷¹, in essence a directive of ensuring healthy, appropriate, and (where possible) rigorous evaluation of claims arising from psychedelic insights. To establish epistemic hygiene, then, is to imbue thoughtful methods or frameworks for 1) demarcating insights and beliefs according to the norms and values of a specific social or cultural context and 2) developing techniques for promoting desired insights. We therefore offer a candidate tool or rubric for thinking through the issue of epistemic hygiene during psychedelic therapy, which may also be highly relevant for any practices or interventions that increase the incidence of insights. Of course, it is worth noting that where each insight ultimately falls is debatable, but the point here is that this veridicality-adaptiveness-falsifiability axis could offer a framework from which psychedelic induced insights can be optimally integrated in clinical settings. We invite future empirical work to shed light on these questions, particularly given forthcoming legalisation (for clinical purposes) of psychedelics in several jurisdictions.

Summary and conclusion

Psychedelics are increasingly considered a viable and effective treatment option for several psychiatric ailments. As such, understanding the mechanisms by which psychedelics confer insight and new beliefs is essential as they become increasingly integrated into clinical settings. However, extant research and theorising has not sufficiently considered the fact that psychedelics, and the feeling of insight they engender, do not necessarily prefer accuracy, and are not necessarily adaptive from a clinical

perspective. This leaves open the possibility that rather than purely offering amelioration, psychedelics may also act as an amplifier for beliefs that enhance existing pathologies or even create new ones. To this end, we have offered the first cohesive account of how psychedelics may confer false beliefs through insight from a neurocomputational perspective – a process we have coined as FIBUS. While we remain optimistic about the future of psychedelic-assisted-therapy, in the interest of averting unfortunate surprises as psychedelic use increases it is important not to overlook the potential for epistemic harm. We also hope that our paper encourages future research on the effects of set, setting, and therapeutic interventions on facilitating valuable insights and adaptive beliefs.

Received: 1 November 2023; Accepted: 23 July 2024;

Published online: 01 August 2024

References

- Ohlsson, S. Restructuring revisited. *Scand. J. Psychol.* **25**, 65–78 (1984).
- Metcalfe, J. Feeling of knowing in memory and problem solving. *J. Exp. Psychol.: Learn., Mem., Cognition* **12**, 288–294 (1986).
- Weisberg, R. W. *Creativity: Understanding innovation in problem solving, science, invention, and the arts*. (John Wiley & Sons Inc, 2006).
- Tulver, K., Kaup, K., Laukkonen, R., Aru, J. Restructuring insight: An integrative review of insight in problem-solving, meditation, psychotherapy, delusions and psychedelics. *Consciousness Cognition* **110**, <https://doi.org/10.31234/osf.io/8fbt9> (2023).
- Laukkonen, R. E. et al. Insight and the selection of ideas. *Neurosci. Biobehav. Rev.* **153**, 105363 (2023).
- Laukkonen, R. E., Kaveladze, B. T., Tangen, J. M. & Schooler, J. W. The dark side of Eureka: Artificially induced Aha moments make facts feel true. *Cognition* **196**, 104122 (2020).
- Laukkonen, R. E. et al. Irrelevant insights make worldviews ring true. *Sci. Rep.* **12**, <https://doi.org/10.1038/s41598-022-05923-3> (2022).
- Grimmer, H., Laukkonen, R., Tangen, J. & von Hippel, W. Eliciting false insights with semantic priming. *Psychonomic Bull. Rev.* **29**, 954–970 (2022).
- Garcia-Romeu, A. et al. Cessation and reduction in alcohol consumption and misuse after psychedelic use. *J. Psychopharmacol.* **33**, 1088–1101 (2019).
- Watts, R., Day, C., Krzanowski, J., Nutt, D. & Carhart-Harris, R. Patients' Accounts of Increased "Connectedness" and "Acceptance" After Psilocybin for Treatment-Resistant Depression. *J. Humanist. Psychol.* **57**, 520–564 (2017).
- Timmermann, C., Watts, R. & Dupuis, D. Towards psychedelic apprenticeship: Developing a gentle touch for the mediation and validation of psychedelic-induced insights and revelations. *Transcultural Psych.* **59**, 691–704 (2022).
- Laukkonen, R. E., Webb, M. E., Salvi, C., Tangen, J. M. & Schooler, J. W. *The Eureka Heuristic: How feelings of insight signal the precision of a new idea*. (2022).
- Luoma, J. B., Chwyl, C., Bathje, G. J., Davis, A. K. & Lancelotta, R. A Meta-Analysis of Placebo-Controlled Trials of Psychedelic-Assisted Therapy. *J. Psychoact. Drugs* **52**, 289–299 (2020).
- McCartney, A. M., McGovern, H. T. & De Foe, A. Vol. 6 10–22 (Akadémiai Kiadó, Hungary, 2022).
- Penn, A., Dorsen, C. G., Hope, S. & Rosa, W. E. Psychedelic-Assisted Therapy: Emerging Treatments in Mental Health Disorders. *Am. J. Nurs.* **121**, 34–40 (2021).
- Maier, N. R. F. Reasoning in humans. II. The solution of a problem and its appearance in consciousness. *J. Comp. Psychol.* **12**, 181–194 (1931).
- Schooler, J. W. & Melcher, J. in *The creative cognition approach*. 97–133 (The MIT Press, 1995).
- Sternberg, R. J. & Davidson, J. E. *The nature of insight*. (The MIT Press, 1995).
- Öllinger, M. & Knoblich, G. in *Recasting Reality: Wolfgang Pauli's Philosophical Ideas and Contemporary Science* (eds Harald Atmanspacher & Hans Primas) 275–300 (Springer Berlin Heidelberg, 2009).
- Ohlsson, S. *Deep learning: How the mind overrides experience*. (Cambridge Univ. Press, 2011).
- Laukkonen, R. E. & Tangen, J. M. How to detect insight moments in problem solving experiments. *Front. Psychol.* **9**, 282 (2018).
- Danek, A. H. & Wiley, J. What about false insights? Deconstructing the Aha! experience along its multiple dimensions for correct and incorrect solutions separately. *Front. Psychol.* **7**, <https://doi.org/10.3389/fpsyg.2016.02077> (2017).
- Jung-Beeman, M. et al. Neural activity when people solve verbal problems with insight (Insight in the brain). *PLoS Biol.* **2**, e97 (2004).
- Topolinski, S. & Reber, R. Gaining insight into the Aha experience. *Curr. Directions Psychol. Sci.* **19**, 402–405 (2010).
- Webb, M. E., Little, D. R. & Cropper, S. J. Insight is not in the problem: Investigating insight in problem solving across task types. *Front. Psychol.* **7**, 1424 (2016).
- Webb, M. E., Little, D. R. & Cropper, S. J. Once more with feeling: Normative data for the aha experience in insight and noninsight problems. *Behav. Res.* **50**, 2035–2056 (2018).
- Danek, A. H., Fraps, T., von Müller, A., Grothe, B. & Öllinger, M. Working wonders? Investigating insight with magic tricks. *Cognition* **130**, 174–185 (2014).
- Hedne, M. R., Norman, E. & Metcalfe, J. Intuitive feelings of warmth and confidence in insight and noninsight problem solving of magic tricks. *Front. Psychol.* **7**, 1314 (2016).
- Laukkonen, R. E., Ingledew, D. J., Grimmer, H. J., Schooler, J. W. & Tangen, J. M. Getting a grip on insight: real-time and embodied Aha experiences predict correct solutions. *Cognition Emot.* **35**, 918–935 (2021).
- Salvi, C., Bricolo, E., Kounios, J., Bowden, E. & Beeman, M. Insight solutions are correct more often than analytic solutions. *Think. Reasoning* **22**, 443–460 (2016).
- Davis, A. K., Barrett, F. S. & Griffiths, R. R. Psychological flexibility mediates the relations between acute psychedelic effects and subjective decreases in depression and anxiety. *J. Contextual Behav. Sci.* **15**, 39–45 (2020).
- Davis, A. K. et al. Development of the Psychological Insight Questionnaire among a sample of people who have consumed psilocybin or LSD. *J. Psychopharmacol.* **35**, 437–446 (2021).
- Gasser, P., Kirchner, K. & Passie, T. LSD-assisted psychotherapy for anxiety associated with a life-threatening disease: a qualitative study of acute and sustained subjective effects. *J. Psychopharmacol.* **29**, 57–68 (2015).
- Schmid, Y., Gasser, P., Oehen, P. & Liechti, M. E. Acute subjective effects in LSD- and MDMA-assisted psychotherapy. *J. Psychopharmacol.* **35**, 362–374 (2021).
- Bogenschutz, M. P. & Pommy, J. M. Therapeutic mechanisms of classic hallucinogens in the treatment of addictions: from indirect evidence to testable hypotheses. *Drug Test. Anal.* **4**, 543–555 (2012).
- Yaden, D. B. & Griffiths, R. R. The Subjective Effects of Psychedelics Are Necessary for Their Enduring Therapeutic Effects. *ACS Pharm. Transl. Sci.* **4**, 568–572 (2021).
- Carbonaro, T. M., Johnson, M. W. & Griffiths, R. R. Subjective features of the psilocybin experience that may account for its self-administration by humans: a double-blind comparison of psilocybin and dextromethorphan. *Psychopharmacol. (Berl.)* **237**, 2293–2304 (2020).
- Carhart-Harris, R. L. & Nutt, D. J. User perceptions of the benefits and harms of hallucinogenic drug use: A web-based questionnaire study. *J. Subst. Use* **15**, 283–300 (2010).

39. Sellers, E. M., Romach, M. K. & Leiderman, D. B. Studies with psychedelic drugs in human volunteers. *Neuropharmacology* **142**, 116–134 (2018).
40. Noorani, T., Garcia-Romeu, A., Swift, T. C., Griffiths, R. R. & Johnson, M. W. Psychedelic therapy for smoking cessation: Qualitative analysis of participant accounts. *J. Psychopharmacol.* **32**, 756–769 (2018).
41. Carhart-Harris, R. L. et al. Psilocybin with psychological support for treatment-resistant depression: six-month follow-up. *Psychopharmacol. (Berl.)* **235**, 399–408 (2018).
42. Williams, M. T. et al. People of color in North America report improvements in racial trauma and mental health symptoms following psychedelic experiences. *Drugs: Educ., Prev. Policy* **28**, 215–226 (2021).
43. Stuyck, H., Aben, B., Cleeremans, A. & Van den Bussche, E. The Aha! moment: Is insight a different form of problem solving? *Conscious. Cognition* **90**, 103055 (2021).
44. Whittlesea, B. W. A. Illusions of familiarity. *J. Exp. Psychol.: Learn., Mem., Cognition* **19**, 1235–1253 (1993).
45. Doss, M. K., Bluestone, M. R. & Gallo, D. A. Two mechanisms of constructive recollection: Perceptual recombination and conceptual fluency. *J. Exp. Psychol. Learn Mem. Cogn.* **42**, 1747–1758 (2016).
46. Doss, M. K., Picart, J. K. & Gallo, D. A. Creating emotional false recollections: Perceptual recombination and conceptual fluency mechanisms. *Emotion* **20**, 750–760 (2020).
47. Dougal, S. & Schooler, J. Discovery Misattribution: When Solving Is Confused With Remembering. *J. Exp. Psychol. Gen.* **136**, 577–592 (2007).
48. Yonelinas, A. P. The Nature of Recollection and Familiarity: A Review of 30 Years of Research. *J. Mem. Lang.* **46**, 441–517 (2002).
49. Gardiner, J. M. Episodic memory and autonoetic consciousness: a first-person approach. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **356**, 1351–1361 (2001).
50. James, W. *The Varieties of Religious Experience*. (Oxford Univ. Press, 1902).
51. Shanon, B. *The antipodes Of the mind: Charting the Phenomenology of the Ayahuasca Experience*. (Oxford Univ. Press, 2002).
52. Grimmer, H. J., Laukkonen, R. E., Freydenzon, A., von Hippel, W. & Tangen, J. M. Thinking style and psychosis proneness do not predict false insights. *Conscious. Cognition* **104**, 103384 (2022).
53. Grimmer, H. J., Tangen, J. M., Freydenzon, A. & Laukkonen, R. E. The illusion of insight: detailed warnings reduce but do not prevent false “Aha!” moments. *Cogn. Emot.* **37**, 329–338 (2023).
54. Kensinger, E. A. Remembering the Details: Effects of Emotion. *Emot. Rev.* **1**, 99–113 (2009).
55. Madan, C. R., Scott, S. M. E. & Kensinger, E. A. Positive emotion enhances association-memory. *Emotion* **19**, 733–740 (2019).
56. Bohn, A. & Berntsen, D. Pleasantness bias in flashbulb memories: Positive and negative flashbulb memories of the fall of the Berlin Wall among East and West Germans. *Mem. Cognition* **35**, 565–577 (2007).
57. Kensinger, E. A. & Schacter, D. L. When the Red Sox shocked the Yankees: Comparing negative and positive memories. *Psychonomic Bull. Rev.* **13**, 757–763 (2006).
58. Barrett, F. S., Johnson, M. W. & Griffiths, R. R. Validation of the revised Mystical Experience Questionnaire in experimental sessions with psilocybin. *J. Psychopharmacol.* **29**, 1182–1190 (2015).
59. McKay, R. T. & Dennett, D. C. The evolution of misbelief. *Behav. Brain Sci.* **32**, 493–510 (2009). discussion 510–461.
60. Sharot, T. The optimism bias. *Curr. Biol.* **21**, R941–R945 (2011).
61. Mishara, A. & Corlett, P. Are delusions biologically adaptive? Salvaging the doxastic shear pin. *Behav. Brain Sci.* **32**, 530–531 (2009).
62. Hohwy, J. The predictive processing hypothesis. *The Oxford handbook of 4E cognition*, 129–145 (Oxford University Press, Oxford UK, 2018).
63. Friston, K. J. et al. Active inference, curiosity and insight. *Neural Comput.* **29**, 2633–2683 (2017).
64. Friston, K. The free-energy principle: a unified brain theory? *Nat. Rev. Neurosci.* **11**, 127–138 (2010).
65. Friston, K. A Free Energy Principle for Biological Systems. *Entropy (Basel)* **14**, 2100–2121 (2012).
66. Bastos, A. M. et al. Canonical microcircuits for predictive coding. *Neuron* **76**, 695–711 (2012).
67. Friston, K. Hierarchical Models in the Brain. *PLOS Comput. Biol.* **4**, e1000211 (2008).
68. Taylor, P., Hobbs, J. N., Burrone, J. & Siegelmann, H. T. The global landscape of cognition: hierarchical aggregation as an organizational principle of human cortical networks and functions. *Sci. Rep.* **5**, 18112 (2015).
69. Vidaurre, D., Smith, S. M. & Woolrich, M. W. Brain network dynamics are hierarchically organized in time. *Proc. Natl. Acad. Sci.* **114**, 12827–12832 (2017).
70. Wacongne, C. et al. Evidence for a hierarchy of predictions and prediction errors in human cortex. *Proc. Natl. Acad. Sci.* **108**, 20754–20759 (2011).
71. Rockland, K. S. & Pandya, D. N. Laminar origins and terminations of cortical connections of the occipital lobe in the rhesus monkey. *Brain Res.* **179**, 3–20 (1979).
72. Murphy, P. C. & Sillito, A. M. Corticofugal feedback influences the generation of length tuning in the visual pathway. *Nature* **329**, 727–729 (1987).
73. Sherman, S. M. & Guillery, R. W. On the actions that one nerve cell can have on another: Distinguishing “drivers” from “modulators”. *Proc. Natl. Acad. Sci.* **95**, 7121–7126 (1998).
74. Badcock, P. B., Friston, K. J. & Ramstead, M. J. D. The hierarchically mechanistic mind: A free-energy formulation of the human psyche. *Phys. Life Rev.* **31**, 104–121 (2019).
75. Keller, G. B. & Msrsc-Flogel, T. D. Predictive Processing: A Canonical Cortical Computation. *Neuron* **100**, 424–435 (2018).
76. Corlett, P. et al. Illusions and delusions: relating experimentally-induced false memories to anomalous experiences and ideas. *Frontiers in Behavioral Neuroscience* **3**, <https://doi.org/10.3389/neuro.08.053.2009> (2009).
77. Hohwy, J. The Self-Evidencing Brain. *Noûs* **50**, 259–285 (2016).
78. Hohwy, J. *The predictive mind*. (OUP Oxford, 2013).
79. Friston, K. & Kiebel, S. Predictive coding under the free-energy principle. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **364**, 1211–1221 (2009).
80. Hobson, J. A. & Friston, K. J. Waking and dreaming consciousness: neurobiological and functional considerations. *Prog. Neurobiol.* **98**, 82–98 (2012).
81. Laukkonen, R. E. & Slagter, H. A. From many to (n)one: Meditation and the plasticity of the predictive mind. *Neurosci. Biobehav. Rev.* **128**, 199–217 (2021).
82. Danek, A. H., Fraps, T., von Müller, A., Grothe, B. & Öllinger, M. Aha! experiences leave a mark: facilitated recall of insight solutions. *Psychol. Res.* **77**, 659–669 (2013).
83. Schwartenbeck, P., FitzGerald, T. H. B. & Dolan, R. Neural signals encoding shifts in beliefs. *NeuroImage* **125**, 578–586 (2016).
84. Mai, X. Q., Luo, J., Wu, J. H. & Luo, Y. J. “Aha!” effects in a guessing riddle task: an event-related potential study. *Hum. Brain Mapp.* **22**, 261–270 (2004).
85. Qiu, J. et al. Brain mechanism of cognitive conflict in a guessing Chinese logograph task. *Neuroreport* **17**, 679–682 (2006).
86. Hodapp, A. & Rabovsky, M. The N400 ERP component reflects an error-based implicit learning signal during language comprehension. *Eur. J. Neurosci.* **54**, 7125–7140 (2021).
87. Friston, K. et al. The anatomy of choice: Dopamine and decision-making. *Philosophical Transactions of The Royal Society B*

- Biological Sciences* **369**, <https://doi.org/10.1098/rstb.2013.0481> (2014).
88. FitzGerald, T. H. B., Dolan, R. J. & Friston, K. Dopamine, reward learning, and active inference. *Front. Comput. Neurosci.* **9**, <https://doi.org/10.3389/fncom.2015.00136> (2015).
89. Haarsma, J. et al. Precision weighting of cortical unsigned prediction error signals benefits learning, is mediated by dopamine, and is impaired in psychosis. *Mol. Psychiatry* **26**, 5320–5333 (2021).
90. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P. & Pezzulo, G. Active inference and learning. *Neurosci. Biobehav. Rev.* **68**, 862–879 (2016).
91. Cristofori, I., Salvi, C., Beeman, M. & Grafman, J. The effects of expected reward on creative problem solving. *Cogn. Affect Behav. Neurosci.* **18**, 925–931 (2018).
92. Oh, Y., Chesebrough, C., Erickson, B., Zhang, F. & Kounios, J. An insight-related neural reward signal. *Neuroimage* **214**, 116757 (2020).
93. Salvi, C., Beeman, M., Bikson, M., McKinley, R. & Grafman, J. TDCS to the right anterior temporal lobe facilitates insight problem-solving. *Sci. Rep.* **10**, 946 (2020).
94. Duzkiewicz, A. J., McNamara, C. G., Takeuchi, T. & Genzel, L. Novelty and Dopaminergic Modulation of Memory Persistence: A Tale of Two Systems. *Trends Neurosci.* **42**, 102–114 (2019).
95. Evans, J. S. B. T. In two minds: dual-process accounts of reasoning. *Trends Cogn. Sci.* **7**, 454–459 (2003).
96. Macchi, L. & Bagassi, M. Intuitive and analytical processes in insight problem solving: a psycho-rhetorical approach to the study of reasoning. *Mind Soc.* **11**, 53–67 (2012).
97. Chuderski, A., Jastrzębski, J., Kroczyk, B., Kucwaj, H. & Ociepka, M. Metacognitive experience on Raven's matrices versus insight problems. *Metacognition Learn.* **16**, 15–35 (2020).
98. Lin, W.-L., Hsu, K.-Y., Chen, H.-C. & Wang, J.-W. The Relations of Gender and Personality Traits on Different Creativities: A Dual-Process Theory Account. *Psychol. Aesthet. Creativity, Arts* **6**, 112–123 (2011).
99. Carhart-Harris, R. L. & Friston, K. J. REBUS and the Anarchic Brain: Toward a Unified Model of the Brain Action of Psychedelics. *Pharm. Rev.* **71**, 316–344 (2019).
100. Vollenweider, F. X. & Preller, K. H. Psychedelic drugs: neurobiology and potential for treatment of psychiatric disorders. *Nat. Rev. Neurosci.* **21**, 611–624 (2020).
101. Preller, K. H. et al. The fabric of meaning and subjective effects in LSD-induced states depend on serotonin 2A receptor activation. *Curr. Biol.* **27**, 451–457 (2017).
102. Mason, N. et al. Me, myself, bye: regional alterations in glutamate and the experience of ego dissolution with psilocybin. *Neuropsychopharmacology* **45**, 2003–2011 (2020).
103. Gattuso, J. J. et al. Default mode network modulation by psychedelics: a systematic review. *Int. J. Neuropsychopharmacol.* **26**, 155–188 (2023).
104. Carhart-Harris, R. L. et al. Neural correlates of the LSD experience revealed by multimodal neuroimaging. *Proc. Natl. Acad. Sci.* **113**, 4853–4858 (2016).
105. de Vos, C. M. H., Mason, N. L. & Kuypers, K. P. C. Psychedelics and Neuroplasticity: A Systematic Review Unraveling the Biological Underpinnings of Psychedelics. *Front Psych.* **12**, 724606 (2021).
106. Beliveau, V. et al. A High-Resolution In Vivo Atlas of the Human Brain's Serotonin System. *J. Neurosci.* **37**, 120–128 (2017).
107. Raichle, M. E. The Brain's Default Mode Network. *Annu. Rev. Neurosci.* **38**, 433–447 (2015).
108. Mars, R. et al. On the relationship between the “default mode network” and the “social brain”. *Front. Human Neurosci.* **6**, <https://doi.org/10.3389/fnhum.2012.00189> (2012).
109. Binder, J. R., Desai, R. H., Graves, W. W. & Conant, L. L. Where is the semantic system? A critical review and meta-analysis of 120 functional neuroimaging studies. *Cereb. Cortex* **19**, 2767–2796 (2009).
110. Alamia, A., Timmermann, C., Nutt, D. J., VanRullen, R. & Carhart-Harris, R. L. DMT alters cortical travelling waves. *Elife* **9**, <https://doi.org/10.7554/eLife.59784> (2020).
111. Timmermann, C. et al. LSD modulates effective connectivity and neural adaptation mechanisms in an auditory oddball paradigm. *Neuropharmacology* **142**, 251–262 (2018).
112. Alamia, A. & VanRullen, R. Alpha oscillations and traveling waves: Signatures of predictive coding? *PLOS Biol.* **17**, e3000487 (2019).
113. Nayak, S. M., Singh, M., Yaden, D. B. & Griffiths, R. R. Belief changes associated with psychedelic use. *J. Psychopharmacol.* **37**, 80–92 (2023).
114. Nayak, S. M. & Griffiths, R. R. A Single Belief-Changing Psychedelic Experience Is Associated With Increased Attribution of Consciousness to Living and Non-living Entities. *Front. Psychol.* **13**, <https://doi.org/10.3389/fpsyg.2022.852248> (2022).
115. Timmermann, C. et al. Psychedelics alter metaphysical beliefs. *Sci. Rep.* **11**, 22166 (2021).
116. Belser, A. B. et al. Patient Experiences of Psilocybin-Assisted Psychotherapy: An Interpretative Phenomenological Analysis. *J. Humanist. Psychol.* **57**, 354–388 (2017).
117. Spitzer, M. et al. Semantic and phonological priming in schizophrenia. *J. Abnorm Psychol.* **103**, 485–494 (1994).
118. Doss, M. K., de Wit, H. & Gallo, D. A. The acute effects of psychoactive drugs on emotional episodic memory encoding, consolidation, and retrieval: A comprehensive review. *Neurosci. Biobehav. Rev.* **150**, 105188 (2023).
119. Kostic, B., Booth, S. E. & Cleary, A. M. The role of analogy in reports of *presque vu*: Does reporting the *presque vu* state signal the near retrieval of a source analogy? *J. Cogn. Psychol.* **27**, 739–754 (2015).
120. Cleary, A. M. et al. Familiarity from the configuration of objects in 3-dimensional space and its relation to *déjà vu*: A virtual reality investigation. *Conscious. Cognition* **21**, 969–975 (2012).
121. Cleary, A. M. & Claxton, A. B. *Déjà Vu*: An Illusion of Prediction. *Psychol. Sci.* **29**, 635–644 (2018).
122. O'Reilly, R. C., Bhattacharyya, R., Howard, M. D. & Ketz, N. Complementary learning systems. *Cogn. Sci.* **38**, 1229–1248 (2014).
123. Voss, J. L., Bridge, D. J., Cohen, N. J. & Walker, J. A. A Closer Look at the Hippocampus and Memory. *Trends Cogn. Sci.* **21**, 577–588 (2017).
124. Singh, D., Norman, K. A. & Schapiro, A. C. A model of autonomous interactions between hippocampus and neocortex driving sleep-dependent memory consolidation. *Proc. Natl. Acad. Sci.* **119**, e2123432119 (2022).
125. Binder, J. R. & Desai, R. H. The neurobiology of semantic memory. *Trends Cogn. Sci.* **15**, 527–536 (2011).
126. Chi, R. P. & Snyder, A. W. Facilitate Insight by Non-Invasive Brain Stimulation. *PLOS ONE* **6**, e16655 (2011).
127. Santarnecchi, E. et al. Gamma tACS over the temporal lobe increases the occurrence of *Eureka!* moments. *Sci. Rep.* **9**, 5778 (2019).
128. Bowles, B. et al. Impaired familiarity with preserved recollection after anterior temporal-lobe resection that spares the hippocampus. *Proc. Natl. Acad. Sci.* **104**, 16382–16387 (2007).
129. Henson, R. N., Cansino, S., Herron, J. E., Robb, W. G. & Rugg, M. D. A familiarity signal in human anterior medial temporal cortex? *Hippocampus* **13**, 301–304 (2003).
130. Ranganath, C. & Ritchey, M. Two cortical systems for memory-guided behaviour. *Nat. Rev. Neurosci.* **13**, 713–726 (2012).
131. Wang, W.-C., Lazzara, M. M., Ranganath, C., Knight, R. T. & Yonelinas, A. P. The Medial Temporal Lobe Supports Conceptual Implicit Memory. *Neuron* **68**, 835–842 (2010).

132. Wang, W. C., Brashier, N. M., Wing, E. A., Marsh, E. J. & Cabeza, R. On Known Unknowns: Fluency and the Neural Mechanisms of Illusory Truth. *J. Cogn. Neurosci.* **28**, 739–746 (2016).
133. Spiers, H. J., Love, B. C., Le Pelley, M. E., Gibb, C. E. & Murphy, R. A. Anterior temporal lobe tracks the formation of prejudice. *J. Cogn. Neurosci.* **29**, 530–544 (2017).
134. Bombardi, C. Neuronal localization of 5-HT_{2A} receptor immunoreactivity in the rat hippocampal region. *Brain Res. Bull.* **87**, 259–273 (2012).
135. De Filippo, R. et al. Somatostatin interneurons activated by 5-HT(2A) receptor suppress slow oscillations in medial entorhinal cortex. *Elife* **10**, <https://doi.org/10.7554/eLife.66960> (2021).
136. Deng, P. Y. & Lei, S. Serotonin increases GABA release in rat entorhinal cortex by inhibiting interneuron TASK-3 K⁺ channels. *Mol. Cell Neurosci.* **39**, 273–284 (2008).
137. Wyskiel, D. R. & Andrade, R. Serotonin excites hippocampal CA1 GABAergic interneurons at the stratum radiatum-stratum lacunosum moleculare border. *Hippocampus* **26**, 1107–1114 (2016).
138. Carhart-Harris, R. et al. The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs. *Front. Human Neurosci.* **8**, <https://doi.org/10.3389/fnhum.2014.00020> (2014).
139. Mason, N. L. et al. Spontaneous and deliberate creative cognition during and after psilocybin exposure. *Transl. Psychiatry* **11**, 209 (2021).
140. Webb, M. E., Cropper, S. J. & Little, D. R. Aha! is stronger when preceded by a “huh?”: presentation of a solution affects ratings of aha experience conditional on accuracy. *Think. Reasoning* **25**, 324–364 (2019).
141. Correia, P. et al. Transient inhibition and long-term facilitation of locomotion by phasic optogenetic activation of serotonin neurons. *eLife* **6**, <https://doi.org/10.7554/eLife.20975> (2017).
142. Cazettes, F., Reato, D., Morais, J. P., Renart, A. & Mainen, Z. F. Phasic Activation of Dorsal Raphe Serotonergic Neurons Increases Pupil Size. *Curr. Biol.* **31**, 192–197.e194 (2021).
143. Matias, S., Lottem, E., Dugué, G. P. & Mainen, Z. F. Activity patterns of serotonin neurons underlying cognitive flexibility. *Elife* **6**, <https://doi.org/10.7554/eLife.20552> (2017).
144. Feldman, H. & Friston, K. Attention, Uncertainty, and Free-Energy. *Front. Human Neurosci.* **4**, <https://doi.org/10.3389/fnhum.2010.00215> (2010).
145. Baggott, M. J., Coyle, J. R., Erowid, E., Erowid, F. & Robertson, L. C. Abnormal visual experiences in individuals with histories of hallucinogen use: a Web-based questionnaire. *Drug Alcohol Depend.* **114**, 61–67 (2011).
146. McCartney, A. M., McGovern, H. T. & De Foe, A. Predictors of Psychedelic Experience: A Thematic Analysis. *J. Psychoact. Drugs* **55**, 411–419 (2023).
147. Hartogsohn, I. Set and setting, psychedelics and the placebo response: An extra-pharmacological perspective on psychopharmacology. *J. Psychopharmacol.* **30**, 1259–1267 (2016).
148. Doss, M. K. et al. Unique effects of sedatives, dissociatives, psychedelics, stimulants, and cannabinoids on episodic memory: A review and reanalysis of acute drug effects on recollection, familiarity, and metamemory. *Psychol. Rev.* **131**, 523–562 (2024).
149. Lebedev, A. V. et al. LSD-induced entropic brain activity predicts subsequent personality change. *Hum. Brain Mapp.* **37**, 3203–3213 (2016).
150. Preller, K. H. et al. Changes in global and thalamic brain connectivity in LSD-induced altered states of consciousness are attributable to the 5-HT_{2A} receptor. *Elife* **7**, <https://doi.org/10.7554/eLife.35082> (2018).
151. Corlett, P. R. et al. Hallucinations and Strong Priors. *Trends Cogn. Sci.* **23**, 114–127 (2019).
152. Reichert, D. P., Seriès, P. & Storkey, A. J. Charles Bonnet syndrome: evidence for a generative model in the cortex? *PLoS Comput. Biol.* **9**, e1003134 (2013).
153. Corlett, P. R., Honey, G. D. & Fletcher, P. C. Prediction error, ketamine and psychosis: An updated model. *J. Psychopharmacol.* **30**, 1145–1155 (2016).
154. Safran, A. *On the Varieties of Conscious Experiences: Altered Beliefs Under Psychedelics (ALBUS)*. (2020).
155. McGovern, H. T., Leptourgos, P., Hutchinson, B. T. & Corlett, P. R. Do psychedelics change beliefs? *Psychopharmacology* **239**, 1809–1821 (2022).
156. Duerler, P., Schilbach, L., Stämpfli, P., Vollenweider, F. X. & Preller, K. H. LSD-induced increases in social adaptation to opinions similar to one’s own are associated with stimulation of serotonin receptors. *Sci. Rep.* **10**, 12181 (2020).
157. Nour, M. M., Evans, L. & Carhart-Harris, R. L. Psychedelics, Personality and Political Perspectives. *J. Psychoact. Drugs* **49**, 182–191 (2017).
158. Agin-Liebes, G. et al. Participant Reports of Mindfulness, Posttraumatic Growth, and Social Connectedness in Psilocybin-Assisted Group Therapy: An Interpretive Phenomenological Analysis. *J. Humanist. Psychol.* **64**, 564–591 (2024).
159. Roseman, L., Nutt, D. J. & Carhart-Harris, R. L. Quality of Acute Psychedelic Experience Predicts Therapeutic Efficacy of Psilocybin for Treatment-Resistant Depression. *Front. Pharm.* **8**, 974 (2017).
160. Letheby, C. *Philosophy of Psychedelics*. (Oxford University Press, 2021).
161. Hartogsohn, I. The Meaning-Enhancing Properties of Psychedelics and Their Mediator Role in Psychedelic Therapy, Spirituality, and Creativity. *Front. Neurosci.* **12**, <https://doi.org/10.3389/fnins.2018.00129> (2018).
162. Girn, M., Mills, C., Roseman, L., Carhart-Harris, R. L. & Christoff, K. Updating the dynamic framework of thought: Creativity and psychedelics. *NeuroImage* **213**, 116726 (2020).
163. Weiss, S. et al. On the Trail of Creativity: Dimensionality of Divergent Thinking and its Relation with Cognitive Abilities, Personality, and Insight. *Eur. J. Personal.* **35**, 291–314 (2021).
164. Yaden, D. B. et al. The noetic quality: A multimethod exploratory study. *Psychol. Conscious.: Theory, Res., Pract.* **4**, 54–62 (2017).
165. Garcia-Romeu, A., Griffiths, R. R. & Johnson, M. W. Psilocybin-occasioned mystical experiences in the treatment of tobacco addiction. *Curr. Drug Abuse. Rev.* **7**, 157–164 (2014).
166. DiVito, A. J. & Leger, R. F. Psychedelics as an emerging novel intervention in the treatment of substance use disorder: a review. *Mol. Biol. Rep.* **47**, 9791–9799 (2020).
167. Veissière, S. P. L., Constant, A., Ramstead, M. J. D., Friston, K. J. & Kirmayer, L. J. Thinking through other minds: A variational approach to cognition and culture. *Behav. Brain Sci.* **43**, e90 (2019).
168. Hou, H., Tian, M. & Zhang, H. Positron emission tomography molecular imaging of dopaminergic system in drug addiction. *Anat. Rec. (Hoboken)* **295**, 722–733 (2012).
169. Gandy, S., Bonnelle, V., Jacobs, E. & Luke, D. Psychedelics as potential catalysts of scientific creativity and insight. *Drug Sci., Policy Law* **8**, 20503245221097649 (2022).
170. Eisen, J. L. et al. The Brown Assessment of Beliefs Scale: reliability and validity. *Am. J. Psych.* **155**, 102–108 (1998).
171. Clark, A. What ‘Extended Me’ knows. *Synthese* **192**, 3757–3775 (2015).
172. Elder, J. H. & Sachs, A. J. Psychophysical receptive fields of edge detection mechanisms. *Vis. Res.* **44**, 795–813 (2004).

Author contribution

This authorship statement was informed by the CRediT authorship taxonomy. Contributions for each author are listed below. H.T.M–Conceptualization, Writing – Original Draft, Writing–Review & Editing, Project administration,

visualization. H.J.G–Conceptualization, Writing–Review & Editing, Project administration. M.K.D– Conceptualization, Writing – Original Draft, Writing – Review & Editing. C. T–Writing–Original Draft, Writing – Review & Editing. B.T.H – Conceptualization, Writing–Original Draft, Writing – Review & Editing. A.L–Writing – Review & Editing, Supervision. P.R.C–Supervision. R.E.L–Conceptualization, Writing – Review & Editing, Visualization, Supervision.

Competing interests

M.K.D. is a consultant for VCENNA.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s44271-024-00120-6>.

Correspondence and requests for materials should be addressed to H. T. McGovern.

Peer review information *Communications Psychology* thanks Thomas Parr, Hartmut Blank and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Marika Schiffer. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024